ZANG, TIANYUE (Selina)

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EDUCATION

Stanford University, Stanford, United States Sept. 2023-Jun.2025

- Master of Science in ***Electrical Engineering*** GPA**: *3.945/4.0***

Miami University, Oxford, United States Aug. 2019-Aug. 2022

- Bachelor of Science in Engineering (***Electrical Engineering*** & ***Computer Engineering***)
- Bachelor of Science in ***Mathematics & Statistics*** & Bachelor of Arts in ***Psychology*** GPA: ***3.92/4.0***

Awards: Magna Cum Laude (08/2022), President's List (Fall Semester 2021-22, Summer Term 2020-21, Spring Semester 2020-21, Fall Semester 2020-21), Dean's List (Spring Semester 2021-22, Spring Semester 2019-20), RedHawk Scholarship for three consecutive years(50% Tuition Grant)

SKILLSET

Computer Language: Python, C++, C, Java, JavaScript, R Language, SQL, VHDL, Verilog, Assembly, Julia, LaTeX

Computer Software and Tools: Mars, MATLAB, SPSS, ANSYS, RStudio, Eclipse, Android Studio, VSCode, Xcode, Vim, Jupyter Notebook, Overleaf, TensorFlow, PyTorch, Scikit-learn, Open3D, MeshLab, MySQL, AWS, CVXPY, Gurobi, Git

Standardized Test: GRE 336(V166 Q170 AW4.5)

PUBLICATION

- Ravi P, **Zang T**, Scott M, Wang M. UAV assisted communication for ground users using machine learning and optimization Journal of Smart Green Computing. 2022; 2(4):175-91. http://dx.doi.org/10.20517/jsegc.2022.05
- M. Wang, R. Zhang and **T. Zang**, "Spatio-Temporal Coordinated Mobile Electric Vehicle Charging in Integrated Transportation and Distribution Systems," *2024 IEEE 99th Vehicular Technology Conference (VTC2024-Spring)*, Singapore, Singapore, 2024, pp. 1-6, doi: 10.1109/VTC2024-Spring62846.2024.10683138.

RESEARCH EXPERIENCE

Psychological Impacts of Excessive Internet Use Stanford, US

Research Member, Supervisor: Professor Johannes Eichstaedt, Stanford University Sept. 2024-Present

- Collaborate with Dr Christopher Kelly to develop a recommender system framework that balances user engagement with mental health, mainly targeting the psychological impacts of excessive internet use;
- Select and justify suitable algorithms, namely ALS for content exploitation and VAE for exploration;
- Utilize reinforcement learning to build two controlled recommender systems: one focused solely on increasing user clicks and the other encoding balancing behaviors to reduce aberrant actions;
- Design experiments involving human subjects to test the efficiency of recommendation systems.

Convex Optimization Applications in Human Activity Analysis Stanford, US

Research Member, Supervisor: Professor Stephen Boyd, Stanford University Apr. 2024-Present

- Apply convex optimization methodologies to a diverse array of liberal disciplines centered on human activities, such as experimental design, decision theory, social psychology network analysis, and resource allocation within psychological services, aiming to improve operational efficiency and effectiveness;
- Utilize tools like CVXPY and Gurobi to address convex optimization challenges;
- Implement optimization techniques to allocate participants in experimental research efficiently, ensuring balanced and valid group compositions to control for confounding variables;
- Improve problem-solving processes by incorporating relaxation and de-symmetrization methods, as well as leveraging the CBC solver for accelerated results;
- Conduct simulations and validations to underscore the practicality and benefits of employing convex optimization.

Impact of Personalized Recommender Systems on Mental Health Stanford, US

Research Member, Supervisor: Professor Ayfer Özgür, Stanford University Apr. 2024-Present

- Conduct an in-depth study of recommender systems, focusing on both traditional (content-based and collaborative filtering) and state-of-the-art ML-based models;
- Delve into the HRNN-meta model and employed the AWS personalization-v2 model across multiple experiments;
- Devise a model using the personalization-v2 model to gather and integrate feedback into a personalized mood prediction model, and implemented a system where if negative mood predictions surpass a set threshold, the personalization model is alternated to enhance mood outcomes;
- Investigate the ethical implications of exploration versus exploitation within contemporary models, highlighting their impact on population impatience and general mental health;
- Explore the correlation between phone usage behavior and mood using ML-based recommender systems for personalized real-time recommendations;
- Utilize datasets like MovieLens to analyze user interactions and moods, demonstrating that sequence significantly impacts outcomes in ML models, unlike frequency—a distinction from non-ML models;
- Associate movie genres with fundamental human moods, identifying noteworthy correlations for 82.92% of users between action intensity and average episode ratings;

- Improve mood detection precision by mapping genres to six basic moods, increasing results to 46.15% (68.24% with a significance threshold of 0.2).

Research on CNN-Based Model Accuracy Enhancement Stanford, US
Independent Research, Supervisor: Professor Ayfer Özgür, Stanford University Jan.-Mar. 2024

- The primary objective of this research was to elevate the accuracy of CNN-based models by focusing on error analysis of earlier predictions;
- Leveraged prior experience with LSTM models to optimize model performance across different scenarios;
- Delved into advanced machine learning techniques, including fine-tuning, semi-supervised learning, two-phase training, and hard example mining;
- Pioneered a novel binary classification approach, resulting in a remarkable 76% accuracy enhancement.

Automated 3D Object Generation with Large Language Models and Diffusion Models Stanford, US
Research Member, Supervisor: Professor Shuran Song, Stanford University Sept.-Dec. 2023

- Investigated methods to automate the generation of large 3D asset datasets for applications in robotics and gaming, addressing a significant scarcity of such assets in the field;
- Utilized Large Language Models (LLMs), specifically Microsoft ' s LLaMA-7B, to generate textual descriptions for 3D object generation, managing the challenge of controlling output structure and length;
- Explored the use of diffusion models like DreamFusion, ProlificDreamer, and TextMesh to transform text inputs into 3D meshes, with a focus on enhancing fidelity and spatial accuracy in the generated objects;
- Conducted extensive experiments to optimize hyperparameters for these models, ensuring high-quality output despite the complexity of 3D object generation;
- Developed an efficient pipeline to streamline the accessibility of these models for others, fostering collaboration and the creation of a diverse and high-quality 3D asset database.

Range Modulation Strategy for Minimizing Interference in Vehicle-to-Vehicle Safety Communication Oxford, US
Research Member, Supervisor: Assistant Prof. Miao Wang, Miami University Feb. 2022-Feb. 2023

- Proposed a digital-twin-based cloud-centric network architecture that would allow vehicles to share more information and enable adaptive modulation of the transmission range of the vehicle-to-vehicle communications, to achieve a satisfactory balance between delay and interference when vehicle disseminates an emergency safety message(ESM) in the event of a sudden, unpredictable emergency on the road;
- Utilized cloud services to construct a digital mock-up of the physical system from collected sensor data, in this way, the reaction time of drivers can be modeled, predicted, and shared with other vehicles;
- Improved safety in the event of unsafe driving conditions by tailoring safety message broadcast range specifically to the present needs of the nearby vehicles using real-time data of neighboring vehicles;
- Drew a conclusion that the ideal transmission range is that which minimizes the amount of interference in the network.

UAV Assisted Communication for Ground Users using Machine Learning and Optimization Oxford, US
Research Member, Supervisor: Assistant Prof. Miao Wang, Miami University May 2021-Feb. 2023

- Proposed to employ solar-powered UAVs with longer flight endurance to provide wireless connectivity to the ground users in rural and remote areas;
- Developed a novel resource allocation system using UAVs to support ground user communications, including getting more reliable connectivity and meeting the Quality of Service (QoS) requirements;
- Analyzed ground users ' densities according to the vehicle location datasets and innovatively split the entire region into grids;
- Determined the vehicle densities in each grid using the proposed partitioning algorithm, and predicted the future densities using Long short-term memory (LSTM) neural networks;
- Devised the linear optimization algorithm to obtain the minimum number of UAVs to support ground user communications based on the anticipated ground users;
- The simulation results demonstrated that the proposed algorithm has increased the signal strength by 50% to provide better connectivity to the ground users.

Automated Vehicle Navigation with Driver Preference Prediction by Machine Learning Oxford, US
Research Member, Supervisor: Assistant Prof. Miao Wang, Miami University May 2021-May 2022

- Developed a novel approach to find the preferred path of the interested driver to fill the gap between current navigation algorithms and autopilot goals, integrating the human driver's preference;
- Extracted the predicted preferred path locations and compared over the minimum density path (MDP) based on the computed traffic density data;
- Built a differential LSTM model with a series of filters to reduce the prediction loss drastically and utilized root-mean-square-error (RMSE) of both latitude and longitude as estimations of optimization to reduce the error below 20 meters for both parameters;
- The simulated results demonstrated that the driver's preferred path outperforms the MDP in finding the shortest and fastest routes, and the preferred path methodology proved that drivers ' preferences can be included in existing algorithms to improve their performance in discovering optimal paths.